

Project Selection Methods

SECTION: Why Project Selection Matters

Organizations typically have more potential projects than resources to execute them. Project selection methods provide an objective (or semi-objective) basis for choosing which projects to fund, comparing financial return, strategic fit, and risk.

SECTION: Net Present Value (NPV)

NPV discounts future cash flows to today's value using a required rate of return (discount rate), then subtracts the initial investment.

Formula: $NPV = \sum [CF_t / (1 + r)^t] - \text{Initial Investment}$

Where CF_t = cash flow in year t , r = discount rate, t = year number.

Decision rule: Accept a project if $NPV > 0$ (it adds value beyond the required rate of return). When comparing mutually exclusive projects, prefer the one with the **higher NPV**.

Worked example: Initial investment = \$100,000. Expected cash flows: Year 1 = \$40,000, Year 2 = \$40,000, Year 3 = \$40,000. Discount rate = 10%.

$$NPV = 40,000/1.1 + 40,000/1.1^2 + 40,000/1.1^3 - 100,000$$

$= 36,364 + 33,058 + 30,053 - 100,000 \approx -\525 (slightly negative — marginally reject at this discount rate).

SECTION: Return on Investment (ROI)

$ROI = (\text{Total discounted benefits} - \text{Total discounted costs}) / \text{Total discounted costs}$, expressed as a percentage.

Higher ROI is preferred. ROI is useful for comparing projects of different sizes proportionally, though it doesn't show absolute value like NPV does.

SECTION: Payback Period

The length of time required to recover the initial investment from net cash inflows.

Formula (simple, even cash flows): $\text{Payback Period} = \text{Initial Investment} / \text{Annual Cash Inflow}$

Worked example: Initial investment = \$90,000, even annual cash inflow = \$30,000/year → $\text{Payback Period} = 90,000/30,000 = 3 \text{ years}$.

For uneven cash flows, subtract each year's cash flow cumulatively from the initial investment until the remaining balance reaches zero.

Limitation: Payback period ignores the time value of money and ignores cash flows after the payback point — it's a liquidity/risk measure, not a profitability measure. Organizations often prefer projects with **shorter** payback periods, especially under capital constraints.

SECTION: Internal Rate of Return (IRR)

The discount rate at which NPV = 0. A project is acceptable if IRR exceeds the organization's required rate of return (hurdle rate/cost of capital). Higher IRR is generally preferred when comparing projects, though IRR can be misleading with non-conventional cash flows (multiple sign changes).

SECTION: Weighted Scoring Model

A technique for comparing projects against multiple criteria (which may include non-financial, strategic factors) using assigned weights.

Method:

1. Identify criteria relevant to the decision (e.g. strategic fit, cost, risk, time to market, customer impact).
2. Assign a weight to each criterion (weights sum to 100% or 1.0).
3. Score each project against each criterion (e.g. 1–10 scale).
4. Multiply each score by its weight and sum for a total weighted score per project.
5. Select the project(s) with the highest weighted score.

Worked example:

Criterion	Weight	Project A Score	Project B Score
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Strategic Fit	0.4	8	6
Cost Efficiency	0.3	5	9
Risk Level	0.3	7	6

Project A weighted score = $0.4(8) + 0.3(5) + 0.3(7) = 3.2 + 1.5 + 2.1 = \mathbf{6.8}$

Project B weighted score = $0.4(6) + 0.3(9) + 0.3(6) = 2.4 + 2.7 + 1.8 = \mathbf{6.9}$

→ Project B is marginally preferred despite a lower strategic-fit score, because it scores much better on cost.

SECTION: Choosing a Method

- **NPV**: best overall measure of absolute value creation; accounts for time value of money.
- **IRR**: useful for ranking by rate of return, intuitive for stakeholders, but can be unreliable with unconventional cash flows.
- **Payback Period**: simple, favors liquidity/quick recovery, ignores time value of money and long-term returns.
- **Weighted Scoring Model**: best when strategic/qualitative factors matter alongside or instead of pure financials.

SECTION: Common Exam Angles

- Calculate NPV, ROI, or Payback Period from given cash flow data.

- Recommend which project to select given competing NPV/IRR/Payback results, and justify with reasoning (not just the number).
- Build/interpret a weighted scoring model table.
- Explain limitations of Payback Period vs NPV.